Q-OGA · Quality-Aligned & Range-Adaptive OGA

Gaussian BEV alternative · classification quality from decoded boxes · five adaptive tasks, without a separate IQA head

Quality-coupled heatmap + range-weighted regression + smooth geometry

Prediction + ground truth · heatmap, metric offset, log-size, doubled-yaw vector

Detached quality map q · no gradient into regression
Default mgiou: clamp(mean 4-axis GIoU, 0, 1)
Alternative rotated_iou: exact aligned polygon IoU
Optional rotated_iou curriculum: a=clip(epoch/Ewarm, 0, 1)
qpeak=(1−a)+a IoU; default Ewarm=0

Range-adaptive weights from GT
world = cell-center grid + GT offset
r = Euclidean distance to origin
wr = 1 + 1.5 clip(r/70.4, 0, 1.5)^2

Heatmap classification
CQFL(p, Y, q)
all grid cells

Offset regression
Range-weighted Smooth-L1
positive cells

Log-size regression
Range-weighted Smooth-L1
positive cells

Doubled-yaw regression
Range-weighted Smooth-L1
positive cells

Oriented geometry
Lgeo = Lproj + βcorner D
derived from regression

T-SBUW · EMA-calibrated five-task objective
Lcls + Loffset + Lsize + Lyaw + Lgeo with learned task weights

CQFL peak target = q^βq; Gaussian neighbors keep negative targets. βq=1, focal power=2.

Geometry supervision · positive cells only

Decode predicted and GT corners
size = exp(clamp(log-size, −10, 10))
angle = 0.5 atan2(sin2, cos2)
center = metric offset

Four projection axes · pred + GT
Lproj = mean[(1 − meanaxis GIoU)/2]
Corner distances: direct d0, reversed d180
Normalize by GT diagonal

Dsoft = mean[−tau logsumexp(−d0/tau, −d180/tau) / diagGT]

tau=0.05 · βcorner=1 · no extra geometry head

Grouped objective

Car
Lcar

Ped / Cyc
Lvru

Ltotal = 0.5 Lcar + 0.5 Lvru

Coordinate loss = meanpositive[wr × sumcoordinate Smooth-L1] · range weighting applies to offset, size and yaw.

Default mgiou quality is a projection proxy; rotated_iou is area IoU. Curriculum epoch starts at zero.